Make Your Science Gateway Discoverable: A Practical Tutorial on AI and Search Optimization

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***Abstract*—As artificial intelligence (AI) systems increasingly mediate access to scientific information, ensuring that Science Gateways are discoverable by both human users and AI interfaces has become a critical priority. This tutorial introduces the emerging discipline of Artificial Intelligence Optimization (AIO), in conjunction with updated search optimization methodologies, to improve the visibility and accessibility of Science Gateways across AI-driven and academic discovery platforms. Attendees will explore methods for identifying prompt patterns that influence inclusion in AI-generated results, applying content clarity techniques to reduce misinterpretation by large language models (LLMs), and structuring content for optimized retrieval by both AI systems and traditional search algorithms. Drawing on case studies and domain expertise, Noreen Whysel and Shari Thurow of the Information Architecture Gateway will highlight frequent AIO challenges and demonstrate best practices. This session is intended for Science Gateways researchers and developers who are looking to enhance usability, credibility, and overall scientific reach.**

***Index Terms*—Artificial intelligence optimization, Science Gateways, prompt patterns, AIO, search engine optimization, SEO, content clarity, large language models, search algorithms**

I. Introduction

As scientific research increasingly relies on web-based infrastructures, Science Gateways play a vital role in providing access to data, tools, and collaboration environments. However, their visibility within both traditional search engines and AI–driven systems remains uneven. In the current digital environment, the discoverability of human users and AI systems - particularly those powered by large language models (LLM) - is essential to ensure the findability, accessibility, usability, and impact of gateway content.

This tutorial explores the emerging field of Artificial Intelligence Optimization (AIO) as it applies to Science Gateways, alongside contemporary search engine optimization (SEO) practices. We present practical strategies for improving content structure, clarity, context, and metadata in ways that align with how AI systems interpret and retrieve information. By identifying prompt patterns and minimizing content ambiguity, developers can enhance the likelihood that gateway resources are correctly cited, summarized, or surfaced in AI-generated outputs.

II. Tutorial Sections

A. Structure and Readability

Effective content design uses clear headings, concise paragraphs, and structured formats—including accessible formats—to enhance readability and guide users through complex technical topics. Introductory text provides simple explanations before introducing technical details.

B. Clarity of Language

Effective communication avoids undefined jargon, maintains consistent terminology, and favors the active voice to enhance clarity. Any necessary technical terms are clearly defined within a specified context.

C. Question and Answer Format

Incorporating Frequently Asked Questions (FAQs) as well as Frequently Unasked Questions that reflect common prompt structures (e.g., “What is...?”, “How does... work?”) can improve content accessibility and alignment with user queries. When implemented on structured web pages, semantic HTML elements enhance both human readability and machine interpretability.

D. Semantic and Technical Optimization

To enhance discoverability and accessibility, content should incorporate structured data (e.g., schema.org types like FAQPage, Article, or HowTo), provide descriptive alternative text for visual elements, and apply ARIA landmarks or roles. Contextual and supplemental navigation to related concepts further supports semantic clarity and effective AI parsing.

E. Data and Context

Content should support claims with citations, references, or datasets and clearly indicate contextual relevance such as audience, discipline, or geographic scope.

F. Metadata and Discoverability

Each page should feature a precise topic-specific title, a succinct meta description within 155–160 characters, and a semantically meaningful URL that accurately reflects the scope and purpose of the content. More metadata is needed depending on the type of digital document (or document surrogate) presented to human and technical users.

G. AI-Specific Considerations

Content should be structured to address common user prompts, using clear examples that AI systems can easily quote or summarize. Repeating key nouns instead of vague pronouns enhances clarity and improves machine interpretability.

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